

Coordinate Geometry



Distance and midpoint

- 1 Find the distance between $A(1, 3)$ and $B(7, 11)$.
 - 2 Find the midpoint of the segment joining $(-3, 5)$ and $(9, -1)$.
 - 3 The midpoint of $A(2, -3)$ and $B(x, y)$ is $(5, 1)$. Find the coordinates of B .
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Gradient and equations of lines

- 4 Find the gradient of the line through $(2, -1)$ and $(6, 7)$.
 - 5 Find the equation of the line through $(4, 3)$ with gradient -2 , in the form $y = mx + c$.
 - 6 Find the equation of the line through $(-1, 4)$ and $(3, -4)$, in the form $y = mx + c$.
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Parallel and perpendicular lines

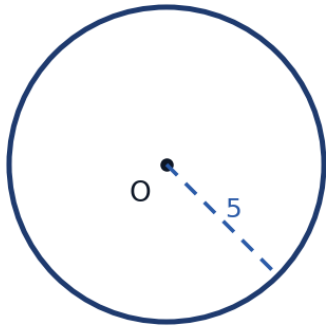
- 7 Find the equation of the line parallel to $y = 3x - 1$ passing through $(5, 0)$.
 - 8 Find the equation of the line perpendicular to $y = \frac{1}{3}x + 2$ passing through $(6, -1)$.
 - 9 Determine whether the lines $y = 4x - 3$ and $4y + x = 8$ are parallel, perpendicular, or neither.
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The equation of a circle

- 10 Write the equation of a circle with center $(3, -2)$ and radius 6.
 - 11 Find the center and radius of $x^2 + y^2 - 4x + 6y - 12 = 0$ by completing the square.
 - 12 A circle has equation $(x + 1)^2 + (y - 4)^2 = 49$. State its center and radius.
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Lines and circles together

- 13 A circle has center $(0, 0)$ and radius 5. Determine whether the line $y = x + 8$ intersects the circle, is tangent to it, or misses it entirely, by comparing the distance from the center to the line against the radius.



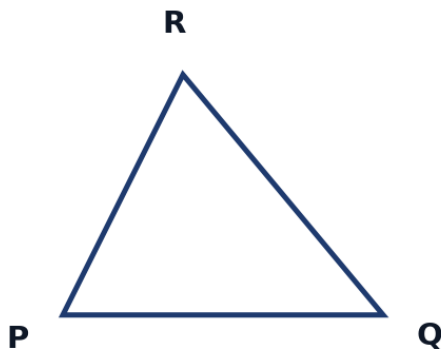
- 14 A circle has center $(2, 1)$ and radius 3. Determine whether the line $x + y = 10$ intersects the circle, is tangent to it, or misses it entirely.

The perpendicular bisector

- 15 Find the equation of the perpendicular bisector of the segment joining $A(1, 2)$ and $B(7, 8)$.
- 16 Find the equation of the perpendicular bisector of the segment joining $A(-2, 3)$ and $B(4, -5)$.

Area of a triangle from coordinates

- 17 Find the area of the triangle with vertices $P(0, 0)$, $Q(8, 0)$, $R(3, 6)$.



- 18 Find the area of the triangle with vertices $A(-3, 2)$, $B(5, 2)$, $C(1, 7)$.

A well-designed word problem: locating a cell tower

- 19** This question has two parts. Two houses are located at $A(0, 0)$ and $B(12, 0)$ (positions in kilometers). A cell tower must be built equidistant from both houses, and it must also lie on the line $y = x + 2$. (a) Find the equation of the perpendicular bisector of AB , and use it together with $y = x + 2$ to find the tower's coordinates. (b) Find the distance from the tower to each house, and confirm they are equal.